

Sec 7.1 – Integration by Substitution

(these notes are adapted from a variety of sources, including notes provided by Dr. Kristin Camenga)

In Chapter 3, we learned rules to differentiate any function obtained by combining constants, powers of x , $\sin x$, $\cos x$, e^x , and $\ln x$, using addition, multiplication, division, or composition of the functions. These functions are called **elementary**.

Now we will consider methods of antidifferentiation. But there is a big difference between looking for derivatives and looking for antiderivatives. Every elementary function has elementary derivatives, but most elementary functions **DO NOT** have elementary antiderivatives.

A good strategy for finding simple antiderivatives is the **guess and check** method. Guess an answer (based on differentiation rules) and then check the answer by differentiating.

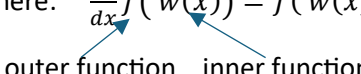
$$\begin{aligned}\frac{d}{dx} e^{x^4} &= e^{x^4} \cdot \frac{d}{dx} x^4 \quad \text{applying the Chain Rule, } e^{\quad} \text{ is the "outer" function and } x^4 \text{ is the inner function} \\ &= e^{x^4} \cdot 4x^3\end{aligned}$$

Our experience with derivatives leads us to notice the relationship between x^4 and $4x^3$. $4x^3$ is just the derivative of x^4 .

So, when we want to find the antiderivative, $\int e^{x^4} \cdot 4x^3 dx$, guess and check would work here, since we recognize the derivative of one function somewhere else in the integrand, from the Chain Rule.

But our goal today is to learn a systematic process for these kind of integrals, where we are undoing the Chain Rule.

We apply the chain rule here: $\frac{d}{dx} f(w(x)) = f(w(x)) \cdot w'(x)$



outer function inner function

If we take the integral of each side: $\int \frac{d}{dx} f(w(x)) dx = \int f(w(x)) \cdot w'(x) dx$

The integral and the derivative on the left-hand side cancel each other out, giving:

$$f(w(x)) = \int f(w(x)) \cdot w'(x) dx$$

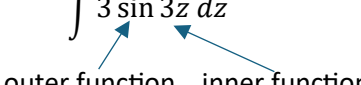
This gives us **Integration by Substitution**, or the **Method of Substitution**. Note that there will be questions on the Final Exam requiring Integration by Substitution.

If $F'(x) = f(x)$, then

$$\int f(w(x)) w'(x) dx = F(w(x)) + C$$

To make a substitution, let w be the "inside function" and $dw = w'(x) dx = \frac{dw}{dx} dx$.

To practice, find these integrals using Integration by Substitution, paying close attention to the inner and outer functions:

$$\int 3 \sin 3z dz$$


outer function inner function

Let $w = 3z$, so $dw = 3 dz$, which gives the substitution: $\int 3 \sin 3z dz = \int \sin w dw = -\cos w + C = -\cos 3z + C$

$$\int t^2 \cdot \sqrt{t^3 + 5} dt$$

(Save the following work on the board, for the upcoming discussion of definite integrals.)

Let $w = t^3 + 5$. (It is a good idea to include constants in your w .)

Then $dw = 3t^2 dt$. But we only have $t^2 dt$ in the integral, so we divide by 3: $\frac{dw}{3} = t^2 dt$

****NOTE:** you can only solve using operations with constants. We cannot have “old” variables in our substituted integral.

Then

$$\begin{aligned} \int t^2 \sqrt{t^3 + 5} dt &= \int \sqrt{w} \frac{dw}{3} = \frac{1}{3} \int w^{1/2} dw \\ &= \frac{1}{3} \frac{w^{3/2}}{3/2} + C = \frac{2}{9} w^{3/2} + C = \frac{2}{9} (t^3 + 5)^{3/2} + C \end{aligned}$$

Partner up

Evaluate the following integrals using substitution:

$$1. \int \frac{5x^4 + 2x}{x^5 + x^2 + 10} dx$$

$$2. \int \sin^2(5x) \cos(5x) dx$$

$$3. \int \tan x dx = \int \frac{\sin x}{\cos x} dx$$

$$4. \int 2x \sin(x^2) e^{\cos(x^2)} dx$$

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1. Let $w = x^5 + x^2 + 10$. Then $dw = (5x^4 + 2x) dx$

$$\begin{aligned}\int \frac{5x^4 + 2x}{x^5 + x^2 + 10} dx &= \int \frac{dw}{w} \\ &= \ln |w| + C = \ln |x^5 + x^2 + 10| + C\end{aligned}$$

2. Let $w = \sin(5x)$ Then $dw = 5 \cos(5x) dx$ or $\frac{dw}{5} = \cos(5x) dx$.

$$\begin{aligned}\int \sin^2(5x) \cos(5x) dx &= \int w^2 \frac{dw}{5} = \frac{1}{5} \int w^2 dw \\ &= \frac{1}{15} w^3 + C = \frac{1}{15} \sin^3(5x) + C\end{aligned}$$

3. Let $w = \cos x$. Then $dw = -\sin x dx$.

$$\begin{aligned}\int \tan x dx &= \int \frac{\sin x}{\cos x} dx = \int \frac{-dw}{w} \\ &= -\ln |w| + C = -\ln |\cos x| + C\end{aligned}$$

NOTE: We could have chosen $w = \sin x$ and had the derivative also in the integral: $dw = \cos x dx$. However, the $\cos x$ is in the denominator of the function and is therefore not multiplied by dx , but rather divides dx . So this substitution would not have worked!

4. Let $w = \cos(x^2)$. Then $dw = -2x \sin(x^2) dx$.

$$\begin{aligned}\int 2x \sin(x^2) e^{\cos(x^2)} dx &= \int -e^w dw \\ &= -e^w + C = -e^{\cos(x^2)} + C\end{aligned}$$

Here we had to recognize that the derivative of our w was a multistep chain rule. One hint is to look at what the 'inside' function is.

Now we would like to evaluate the **definite** integral: $\int_0^1 t^2 \cdot \sqrt{t^3 + 5} dt$

The nice thing is we've already done the work, we just need to include the limits of integration.

Options for evaluating definite integrals using substitution:

- Find the antiderivative using substitution and then evaluate using the FTC and the original limits.
- When doing substitution, also substitute for the limits, switching all terms into the new variable. Then evaluate in the new variable, finding the antiderivative and using the new limits.

$$\int_0^1 t^2 \cdot \sqrt{t^3 + 5} \, dt = \frac{2}{9} (t^3 + 5)^{3/2} \Big|_0^1 = \frac{2}{9} ((1^3 + 5)^{3/2} - (0^3 + 5)^{3/2}) = \frac{2}{9} (6^{3/2} + 5^{3/2}) \approx 5.751$$

$$\int_0^1 t^2 \cdot \sqrt{t^3 + 5} \, dt = \frac{1}{3} \int_5^6 \sqrt{w} \, dw = \frac{1}{3} \cdot \frac{2}{3} w^{3/2} \Big|_5^6 = \frac{2}{9} (6^{3/2} + 5^{3/2}) \approx 5.751$$

Using the substitutions $w = t^3 + 5$, $dw = 3t^2 \, dt$, $\frac{dw}{3} = t^2 \, dt$

Partner up

Evaluate $\int_0^{\pi/4} \sin(x) \cos^4(x) \, dx$

Option 1: Complete antiderivative, then FTC

Let $w = \cos(x)$. Then $dw = -\sin(x) \, dx$ and

$$\int \sin(x) \cos^4(x) \, dx = \int -w^4 \, dw = -\frac{1}{5} w^5 + C = -\frac{1}{5} \cos^5(x) + C$$

$$\begin{aligned} \int_0^{\pi/4} \sin(x) \cos^4(x) \, dx &= -\frac{1}{5} \cos^5(x) \Big|_0^{\pi/4} \\ &= -\frac{1}{5} \cos^5(\pi/4) + \frac{1}{5} \cos^5(0) = -\frac{1}{5} \left(\frac{\sqrt{2}}{2} \right)^5 + \frac{1}{5} \end{aligned}$$

Option 2: Change all variables, including limits, and then do FTC

Let $w = \cos(x)$. Then $dw = -\sin(x) \, dx$, $w(0) = \cos(0) = 1$, and

$$w(\pi/4) = \cos(\pi/4) = \frac{\sqrt{2}}{2}.$$

$$\begin{aligned} \int_0^{\pi/4} \sin(x) \cos^4(x) \, dx &= \int_1^{\sqrt{2}/2} -w^4 \, dw \\ &= -\frac{1}{5} w^5 \Big|_1^{\sqrt{2}/2} = -\frac{1}{5} \left(\frac{\sqrt{2}}{2} \right)^5 + \frac{1}{5} \end{aligned}$$

There are more complex substitutions discussed at the end of the section in the textbook.